

TUTORIAL -3

Creating a Virtual Machine

3/2/26

Bafna Gold
Date: _____ Page: _____

Tutorial 3

1) creating a virtual machine

- VMware Workstation Pro 17 was installed and launched.
- A new virtual machine was created using the custom option.
- Installation media was selected using an ISO file.
- The guest operating system and its version were chosen.
- The virtual machine was configured by allocating RAM, CPU cores and disk space.
- The set up was completed and OS was installed successfully.

2) installing Python on the virtual machine

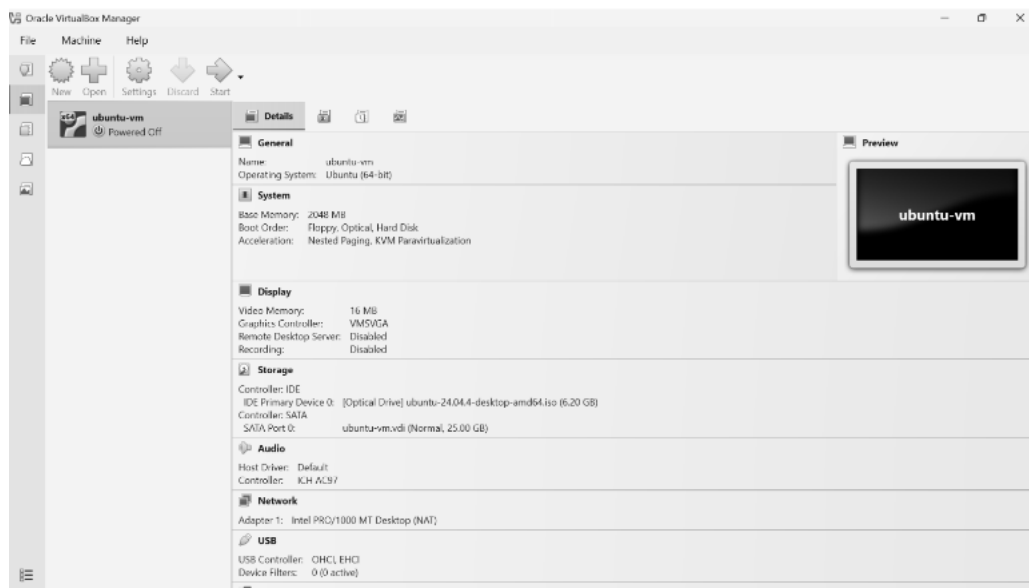
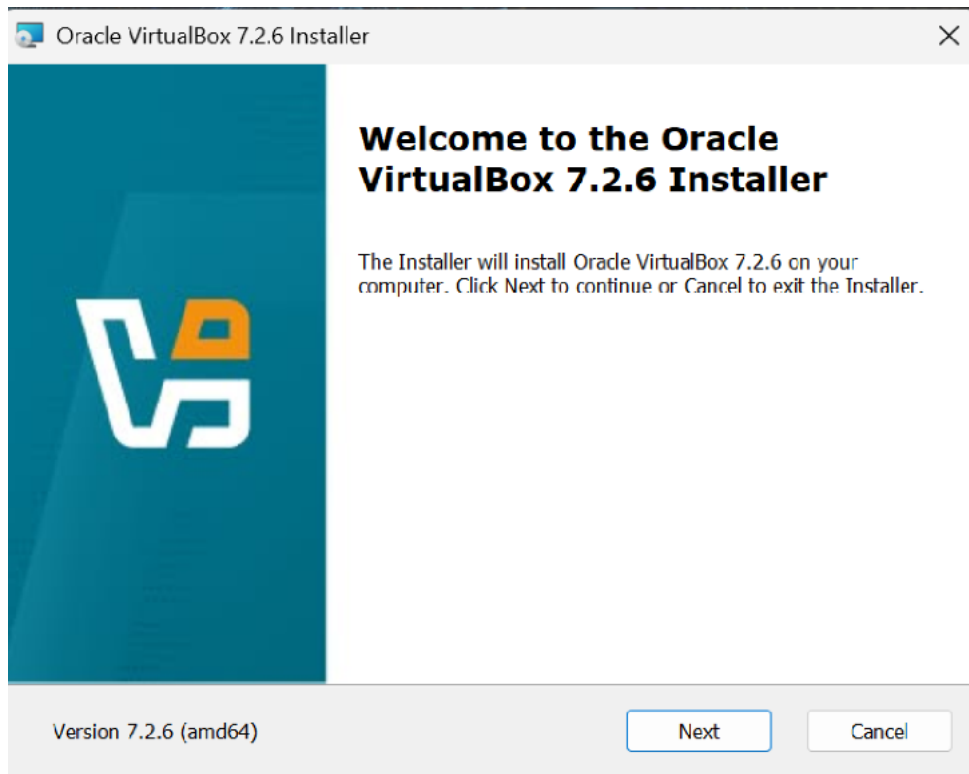
- The VM was started and user logged into the OS.
- For Ubuntu: Python and pip were installed using terminal commands.
- For Windows: Python installer was downloaded.

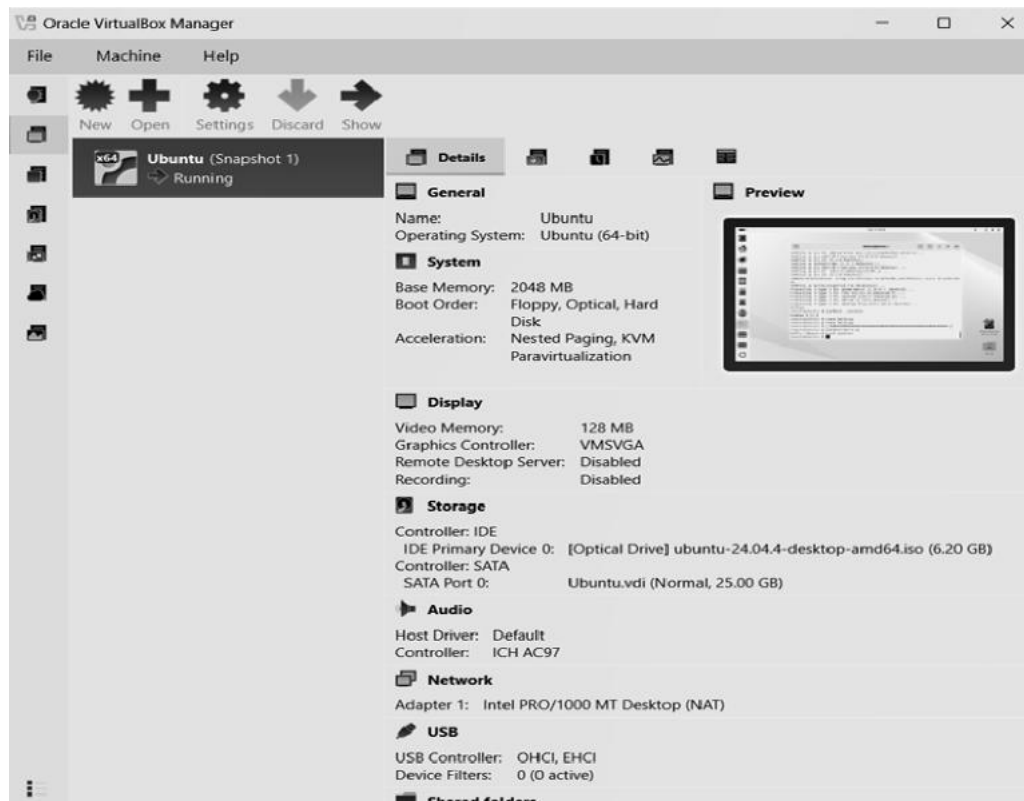
3) Part 1: - Running a Simple Python Program

- A Python script was created to implement a basic machine learning model inside the VM.
- Required libraries such as NumPy, Pandas, Scikit-learn were imported for data handling and model building.
- A dataset was loaded and preprocessed by

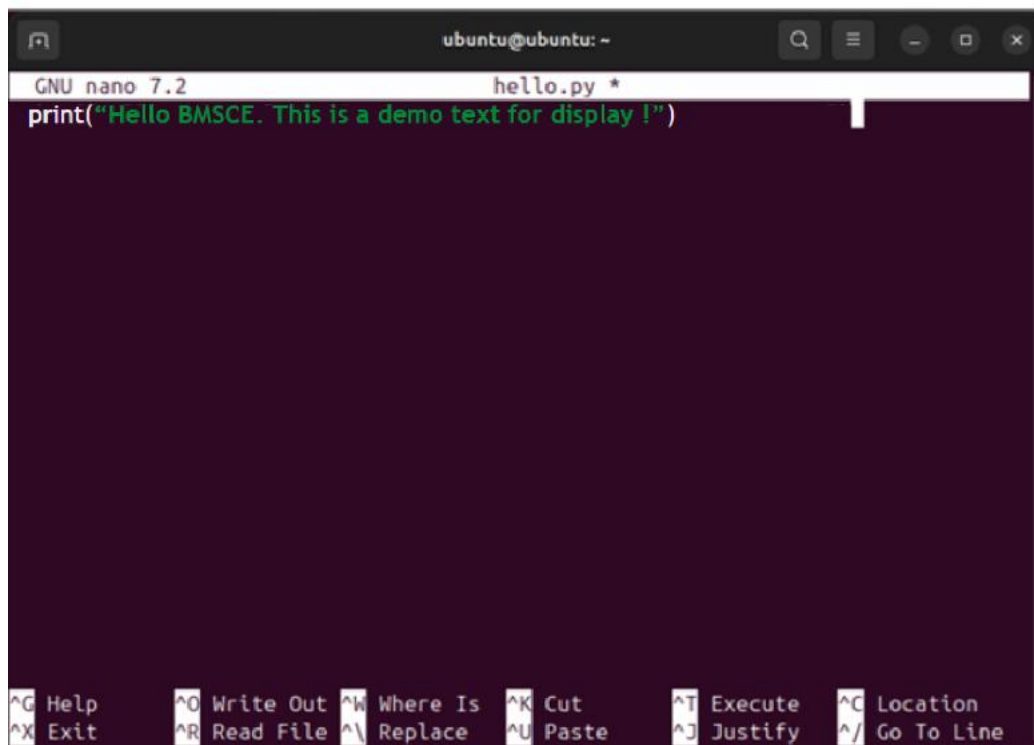
handling missing values and selecting relevant features.

- The dataset was divided into training and testing sets to evaluate the model performance.
- A suitable machine learning algorithm (such as linear regression, logistic regression, or decision tree) was selected and trained using the training data.
- The trained model was then tested using the test dataset to make predictions.
- The performance of the model was evaluated using appropriate metrics such as accuracy, mean squared error, or precision.
- The program was executed successfully, and the output confirmed that the ML model was correctly within the virtual environment.

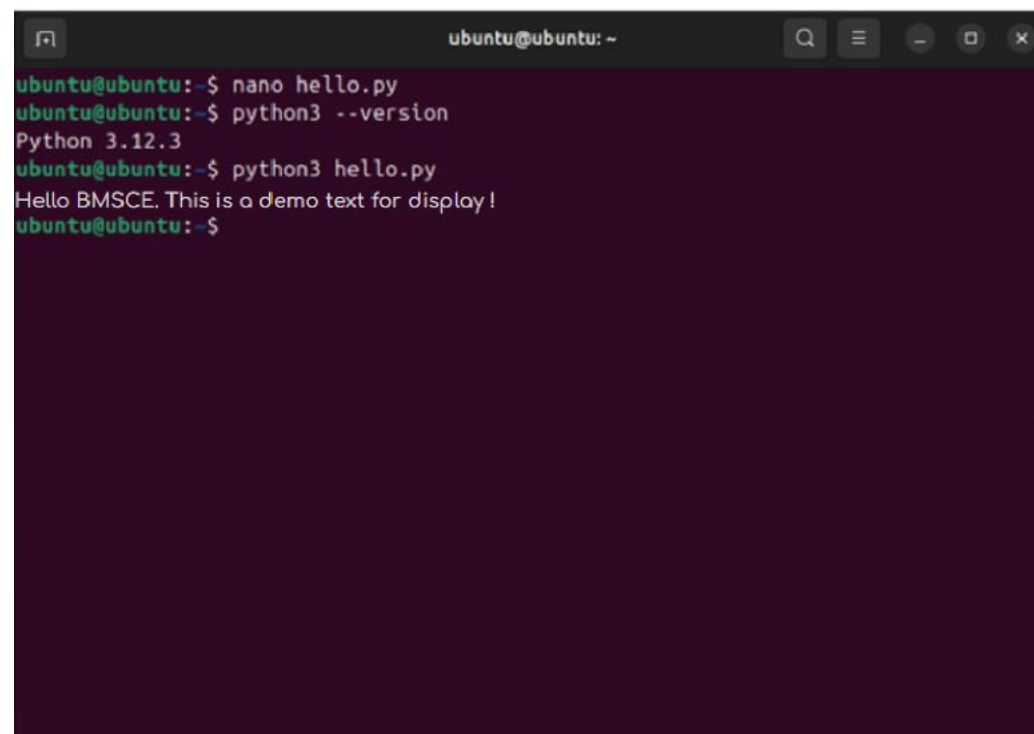




```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ sudo apt update  
Ign:1 cdrom://Ubuntu 24.04.4 LTS _Noble Numbat_ - Release amd64 (20260210) noble  
InRelease  
Hit:2 cdrom://Ubuntu 24.04.4 LTS _Noble Numbat_ - Release amd64 (20260210) noble  
Release  
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1579  
kB]  
Get:6 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [400 k  
B]  
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [253  
kB]  
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21  
.5 kB]  
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata  
[10.7 kB]  
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe i386 Packages [7  
91 kB]  
Hit:11 http://archive.ubuntu.com/ubuntu noble InRelease  
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages  
[1170 kB]  
Get:13 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]  
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en  
[225 kB]  
  
ubuntu@ubuntu:~$ sudo apt install python3 python3-pip -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
python3 is already the newest version (3.12.3-0ubuntu2.1).  
python3 set to manually installed.
```



```
ubuntu@ubuntu: ~  
GNU nano 7.2 hello.py *  
print("Hello BMSCE. This is a demo text for display !")  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```



```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ nano hello.py  
ubuntu@ubuntu:~$ python3 --version  
Python 3.12.3  
ubuntu@ubuntu:~$ python3 hello.py  
Hello BMSCE. This is a demo text for display!  
ubuntu@ubuntu:~$
```